

ing maintaining controls and storage separation of data. With management for up to 100 host systems, you are no longer limited to which system's data can be backed up, or using a dedicated backup server you can quickly and cost-effectively protect all of them," describes Bharat Kumar.

Subram Natarajan of IBM points out that RAID 6 implementations at the hardware level to protect from double disk failures are now in the mainstream. RAID 6 is a redundant array of inexpensive striped disks with dual parity and combines four or more disks in a way that protects data against loss of any two disks.

The next step is a self-healing array, which can eliminate the risk of a single point of failure that exists in RAID arrays. It reduces the risk of data loss on a disk drive by inspecting adjacent areas around the first defect and then reconstructing the data associated with the first defect using parity. IBM's TotalStorage Enterprise Storage Server and EMC's Centra are examples of self-healing systems.

Software-based deduplication is another interesting backup enabler. "Software-based deduplication offers enterprises the ability to store their backup data, anywhere, on any type of storage. Enterprises can keep more backup data on disk in the data centre, and leverage the bandwidth efficiency to replicate the data to a disaster recovery site," says Joshi of Symantec. Backup is a key feature of FileStore solution.

Backup is gaining importance on the personal front as well. While operating systems also come with backup solutions, comprehensive third-party solutions are also available. One example is Conjoin's ClickFree. "It is a pack of devices that comes with a software and makes data transfer an easy task. Clickfree allows the users to backup their data by simply plugging it into the USB port of the computer. It also stores data from multiple computers," says G. Rameshraj, head-sales and marketing, Conjoin Group.

Iomega, an EMC company, has a

Key developments in storage

Personal storage

- Increasing capacity; decreasing size
- More security features in devices ranging from hard drives to thumb drives
- Solid-state devices
- Holistic backup tools
- USB 3.0
- RAID 6

Enterprise storage

- Storage evolves into Information Management
- Deduplication
- Thin provisioning
- Virtualisation
- Cloud storage (storage as a service)
- Virtual data centres
- Fibre Channel over Ethernet, SAS-2, 1 and 10 GB Ethernet and other networking advancements
- Highly developed backup and disaster recovery options
- RAID 6, self-healing arrays
- Green storage

range of products including multimedia, network, portable and desktop drives.

Storage, networking and interfaces

"Storage hardware has evolved in host interfaces, 8 GB/sec Fibre Channel, 6 GB/sec serial-attached SCSI (SAS), and 1 and 10 GB/sec iSCSI," comments Mahadevan of LSI, summing up the key developments on the networking and host interface front.

It is well-known that fibre channels are extremely popular in storage networking. But this popularity is likely to be challenged by SAS and 10 GB Ethernet.

"At 10 GB, Ethernet becomes more of a storage play because it boosts iSCSI's performance. And the serial-attached SCSI (SAS-2) standard will fortify that interface as 'enterprise class,' largely because of its 6 GB/sec capabilities. That means SAS should start to challenge FC for high-performance disk drives," predicts SearchStorage.com

In October, LSI launched what it

claims to be the industry's first 6 GB/sec SAS-to-SATA bridge cards and 16-port SAS storage processors.

Fibre Channel Over Ethernet (FCoE) is another emerging converged network protocol that was finalised in June 2008 by the International Committee for Information Technology Standards' T11 Technical Committee.

FCoE-based products have started hitting the market. This year, Mellanox Technologies released the industry's first 40 GB Ethernet adaptor, the ConnectX-2 EN 40G, with FCoE support. Brocade has released a new model of its DCX Backbone switches to specially handle Fibre Channel traffic between two data centres using Ethernet.

It is interesting to note that a single product can support multiple protocols. LSI's Engenio 4900, for example, has both 8GB/s Fibre Channel (8GFC) and 1GB/s iSCSI host interfaces.

"By utilising the system's 8GFC host interfaces, users can reduce the number of host bus adaptors (HBAs) per server and the number of overall ports in the fibre channel storage area network (FC SAN) infrastructure while achieving comparable performance," says Mahadevan.

On the personal storage front, USB 3.0 is a technology to watch out for. "It has been approved by the USB forum recently. The technology lets people transfer data at a fast speed. Its maximum transfer rate is 10 times that of USB 2.0 and hence is called the SuperSpeed USB," says Sivaswamy. External hard drives such as Buffalo Technology's DriveStation are already available with USB 3.0.

Smart nodes

It is easy to throw away an old thumb drive and start using a new one. But at the enterprise level, adopting a new technology is very challenging. The IT team needs to think about data migration or data loss, application downtime, forklift upgrades, provisioning, etc. The storage industry is trying to address this by offering intelligent devices that can smoothen the transition from old to